

LINUX SWAP & SWAPPINESS

Swap

This page provides an introduction to [swap space and paging](#) on GNU/Linux. It covers creation and activation of swap partitions and swap files.

From [All about Linux swap space](#):

Linux divides its physical RAM (random access memory) into chunks of memory called pages. Swapping is the process whereby a page of memory is copied to the preconfigured space on the hard disk, called swap space, to free up that page of memory. The combined sizes of the physical memory and the swap space is the amount of virtual memory available.

Support for swap is provided by the Linux kernel and user-space utilities from the [util-linux](#) package.

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Swap space

Swap space can take the form of a disk partition or a file. Users may create a swap space during installation or at any later time as desired. Swap space can be used for two purposes, to extend the virtual memory beyond the installed physical memory (RAM), and also for [suspend-to-disk](#) support.

If it is beneficial to extend the virtual memory with swap depends on the amount of installed physical memory. If the amount of physical memory is less than the amount of memory required to run all the desired programs, then it *may be* beneficial to enable swap. This avoids [out of memory conditions](#), where the Linux kernel OOM killer mechanism will automatically attempt to free up memory by killing processes. To increase the amount of virtual memory to the required amount, add the necessary difference (or more) as swap space.

The biggest drawback of enabling swap is its lower performance, see section [#Performance](#). Hence, enabling swap is a matter of personal preference: some prefer programs to be killed over enabling swap and others prefer enabling swap and slower system when the physical memory is exhausted.

Note: There is no performance difference between using a swap partition and a contiguous swap file.

To check swap status, use:

```
$ swapon --show
```

Or to show physical memory as well as swap usage:

```
$ free -h
```

Swap partition



The factual accuracy of this article or section is disputed.



Reason: [82](#) is the swap [partition type](#) on MBR, there is no swap partition autodetection on MBR. / TRIM commands on swap partitions seem to be issued automatically by the Kernel if supported see [Fedora Deployment Guides](#). (Discuss in [Talk:Swap#Clarify swap discovery](#))

A swap partition can be created with most GNU/Linux [partitioning tools](#). Swap partitions are typically designated as type [82](#). Even though it is possible to use any partition type as swap, it is recommended to use type [82](#) in most cases since [systemd](#) will automatically detect it and mount it (see below).

To set up a partition as Linux swap area, the [mkswap\(8\)](#) command is used. For example:

```
# mkswap /dev/sdxy
```

Warning: All data on the specified partition will be lost.

To enable the device for paging:

```
# swapon /dev/sdxy
```

To enable this swap partition on boot, add an entry to [/etc/fstab](#):

```
UUID=device_UUID none swap defaults 0 0
```

where the `device_UUID` is the **UUID** of the swap space.

See **fstab** for the file syntax.

Note:

- The fstab-entry is optional if the swap partition is located on a device using GPT. See the next subsection.
- If using an SSD with **TRIM** support, consider using `discard` in the swap line in **fstab**. If activating swap manually with *swapon*, using the `-d / --discard` parameter achieves the same. See [swapon\(8\)](#) for details.

Warning: Enabling discard on RAID setups using mdadm will cause system lockup on boot and during runtime, if using swapon.

Activation by systemd

systemd activates swap partitions based on two different mechanisms. Both are executables in `/usr/lib/systemd/system-generators`. The generators are run on start-up and create native systemd units for mounts. The first, `systemd-fstab-generator`, reads the fstab to generate units, including a unit for swap. The second, `systemd-gpt-auto-generator` inspects the root disk to generate units. It operates on GPT disks only, and can identify swap partitions by their type GUID, see [systemd#GPT partition automounting](#) for more information.

Disabling swap

To deactivate specific swap space:

```
# swaponoff /dev/sdxy
```

Alternatively use the `-a` switch to deactivate all swap space.

Since swap is managed by systemd, it will be activated again on the next system startup. To disable the automatic activation of detected swap space permanently, run `systemctl --type swap` to find the responsible *.swap* unit and **mask** it.

Swap file

As an alternative to creating an entire partition, a swap file offers the ability to vary its size on-the-fly, and is more easily removed altogether. This may be especially desirable if disk space is at a premium (e.g. a modestly-sized SSD).

Warning: **Btrfs** supports swap file with limitations since Linux kernel version 5.0. See [Btrfs#Swap file](#) for more information.

Manually

Swap file creation

Note: If you use Btrfs, use the procedure described in [Btrfs#Swap file](#) instead of the following *dd* command to initialize the swap file.

Use **dd** to create a swap file the size of your choosing. For example, creating a 512 MiB swap file:

```
# dd if=/dev/zero of=/swapfile bs=1M count=512 status=progress
```

Note: Using *dd* to allocate a swap file is the most portable solution, see [swapon\(8\) § Files with holes](#) for details.

Set the right permissions (a world-readable swap file is a huge local vulnerability):

```
# chmod 600 /swapfile
```

After creating the correctly sized file, format it to swap:

```
# mkswap /swapfile
```

Activate the swap file:

```
# swapon /swapfile
```

Finally, edit the fstab configuration to add an entry for the swap file:

```
/etc/fstab
/swapfile none swap defaults 0 0
```

For additional information, see [fstab#Usage](#).

Note:

- The swap file must be specified by its location on the file system, not by its UUID or LABEL.
- When using **Btrfs**, do not forget to add the created subvolume to the list as well, and remove the `discard`, `autodefrag` and compression options.

Remove swap file

To remove a swap file, it must be turned off first and then can be removed:

```
# swaponoff /swapfile
# rm -f /swapfile
```

Finally remove the relevant entry from `/etc/fstab`.

Automated

systemd-swap

systemd-swap is a script for creating hybrid swap space from zram swaps, swap files and swap partitions. It is not affiliated with the systemd project.

Install the **systemd-swap** package. Uncomment and set `swapfc_enabled=1` in the *Swap File Chunked* section of `/etc/systemd/swap.conf`. **Start/enable** the `systemd-swap` service.

Visit the [authors GitHub](#) page for more information and setting up the **recommended configuration**.

Note:

- If the journal keeps showing the following warning `systemd-swap[.]: WARN: swapFC: ENOSPC` and no swap file is being created, you need to uncomment and set `swapfc_force_preallocated=1` in `/etc/systemd/swap.conf`.
- The swap file created by systemd-swap cannot be easily used for **hibernation**. See [systemd-swap issue 85](#).

Swap encryption

See [dm-crypt/Swap encryption](#).

Performance

Swap operations are usually significantly slower than directly accessing data in RAM. Disabling swap entirely to improve performance can sometimes lead to a degradation, since it decreases the memory available for VFS caches, causing more frequent and costly disk I/O.

Swap values can be adjusted to help performance:

Swappiness

The *swappiness* `sysctl` parameter represents the kernel's preference (or avoidance) of swap space. Swappiness can have a value between 0 and 100, the default value is 60. A low value causes the kernel to avoid swapping, a high value causes the kernel to try to use swap space, and a value of 100 means IO cost is assumed to be equal. Using a low value on sufficient memory is known to improve responsiveness on many systems.

To check the current swappiness value:

```
$ sysctl vm.swappiness
```

Alternatively, the files `/sys/fs/cgroup/memory/memory.swappiness` or `/proc/sys/vm/swappiness` can be read in order to obtain the raw integer value.

Note: As `/proc` is a lot less organized and is kept only for compatibility purposes, you are encouraged to use `/sys` instead.

To temporarily set the swappiness value:

```
# sysctl -w vm.swappiness=10
```

To set the swappiness value permanently, create a `sysctl.d(5)` configuration file. For example:

```
/etc/sysctl.d/99-swappiness.conf
```

```
vm.swappiness=10
```

To test and more on why this may work, take a look at [this article](#).

VFS cache pressure

Another *sysctl* parameter that affects swap performance is `vm.vfs_cache_pressure`, which controls the tendency of the kernel to reclaim the memory which is used for caching of VFS caches, versus pagecache and swap. Increasing this value increases the rate at which VFS caches are reclaimed [\[1\]](#)[\[dead link\]](#) 2020-04-03 ^[1]. For more information, see the [Linux kernel documentation](#).

Priority

If you have more than one swap file or swap partition you should consider assigning a priority value (0 to 32767) for each swap area. The system will use swap areas of higher priority before using swap areas of lower priority. For example, if you have a faster disk (`/dev/sda`) and a slower disk (`/dev/sdb`), assign a higher priority to the swap area located on the fastest device. Priorities can be assigned in `fstab` via the `pri` parameter:

```
/dev/sda1 none swap defaults,pri=100 0 0
/dev/sdb2 none swap defaults,pri=10 0 0
```

Or via the `--priority` parameter of *swapon*:

```
# swapon --priority 100 /dev/sda1
```

If two or more areas have the same priority, and it is the highest priority available, pages are allocated on a round-robin basis between them.

Using zswap or zram

Zswap is a Linux kernel feature providing a compressed write-back cache for swapped pages. This increases the performance and decreases the IO-Operations. **ZRAM** creates a virtual compressed Swap-file in memory as alternative to a swapfile on disk.

Striping

There is no necessity to use **RAID** for swap performance reasons. The kernel itself can stripe swapping on several devices, if you just give them the same priority in the `/etc/fstab` file. Refer to [The Software-RAID HOWTO](#) for details.

Category: [File systems](#)

CHANGE/CHECK SWAPPINESS VALUE ON LINUX

Swappiness is the kernel parameter that defines how much (and how often) your Linux kernel will copy RAM contents to swap. This parameter's default value is "60" and it can take anything from "0" to "100".

The higher the value of the swappiness parameter, the more aggressively your kernel will swap.

Why change it?



The default value is a one-fit-all solution that can't possibly be equally efficient in all of the individual use cases, hardware specifications and user needs. Moreover, the swappiness of a system is a primary factor that determines the overall functionality and speed performance of an OS. That said, it is very important to understand how swappiness works and how the various configurations of this element could improve the operation of your system and thus your everyday usage experience.

As RAM memory is so much larger and cheaper than it used to be in the past, there are many users nowadays that have enough memory to almost never need to use the swap file. The obvious benefit that derives from this is that no system resources are ever occupied by the swapping process and that cached files are not moved back and forth from the RAM to the swap and vice versa for no reason.

#Check current swappiness value:

```
$ cat /proc/sys/vm/swappiness
```

#Change Swappiness Value:

The swappiness parameter value is stored in a simple configuration text file located in `/proc/sys/vm` and is named "swappiness". If you navigate there through the file manager, you will be able to locate the file and open it to check your system's swappiness.

You can also check it or change it through the terminal (which is faster) by typing the following command:

```
$ sudo sysctl vm.swappiness=<value between 0 and 100>
```

```
$ sudo sysctl vm.swappiness=8
```

or whatever else between "0" and "100" instead of the value "8" that I used. To ensure that the swappiness value was correctly changed to the desired one, you simply type: `$ cat /proc/sys/vm/swappiness`

To set the swappiness value **permanently**, create a **sysctl.d** configuration file. For example:

```
/etc/sysctl.d/99-swappiness.conf  
vm.swappiness=8
```

or execute `$ sudo sysctl vm.swappiness=<value between 0 and 100>`
at every boot up via a systemd unit file /systemd service file